AI Assisted Coding

Lab Assignment - 14.5

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Batch No: 02

Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS Generation

Lab Objectives

- Design functional, visually appealing web applications using HTML, CSS, and JavaScript with AI assistance.
- Apply responsive, accessible, and interactive design principles.
- Create practical UI components for real-world web applications.
- Use AI to optimize layout, UX, and performance.

Task 1 – Image Gallery Website

Create an image gallery web page.

Requirements:

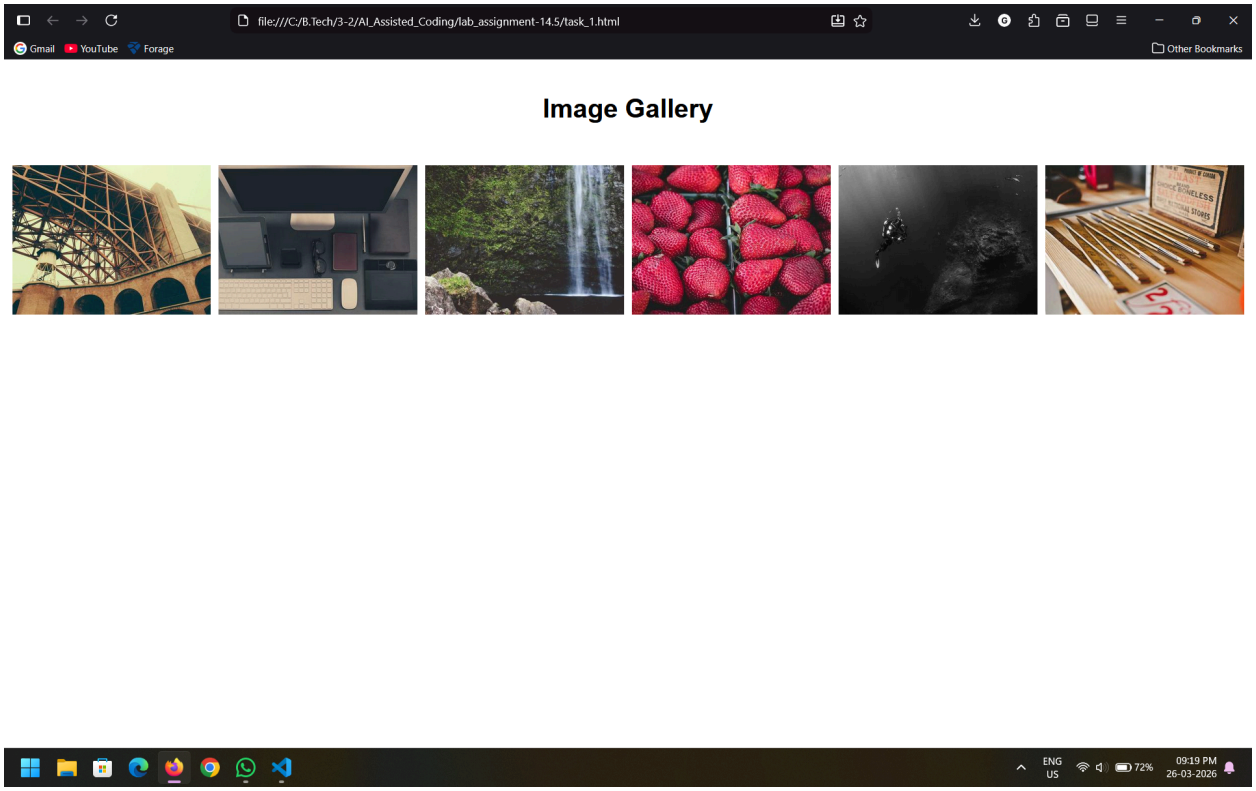
- Display images in a grid layout.
- Use AI to:
    - o Optimize layout using CSS Grid.
    - o Add hover zoom or overlay effects.
    - o Ensure mobile responsiveness.

Copilot Suggested Code :

task_1.html > html

```html
<!— Create an image gallery web page. —>

<!DOCTYPE html>
<html lang="en">
<head>
    <title>Image Gallery</title>
    <style>
        body {
            font-family: 'Arial', sans-serif;
            margin: 0;
            padding: 0;
        }
        .gallery {
            display: grid;
            grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
            gap: 10px;
            padding: 10px;
        }
        .gallery img {
            width: 100%;
            height: auto;
            transition: transform 0.3s ease, box-shadow 0.3s ease;
        }
        .gallery img:hover {
            transform: scale(1.05);
            box-shadow: 0 4px 8px rgba(0,0,0,0.2);
        }
        @media (max-width: 600px) {
            .gallery {
                grid-template-columns: repeat(auto-fit, minmax(150px, 1fr));
            }
        }
    </style>
</head>
<body>
    <h1 style="text-align: center; padding: 20px;">Image Gallery</h1>
    <div class="gallery">
        <img src="https://picsum.photos/seed/1/400/300" alt="Image 1">
        <img src="https://picsum.photos/seed/2/400/300" alt="Image 2">
        <img src="https://picsum.photos/seed/3/400/300" alt="Image 3">
        <img src="https://picsum.photos/seed/4/400/300" alt="Image 4">
        <img src="https://picsum.photos/seed/5/400/300" alt="Image 5">
        <img src="https://picsum.photos/seed/6/400/300" alt="Image 6">
    </div>
</body>
</html>
```

Output :



Task 2 – Profile Card Generator

Create a profile card web page.

Requirements:

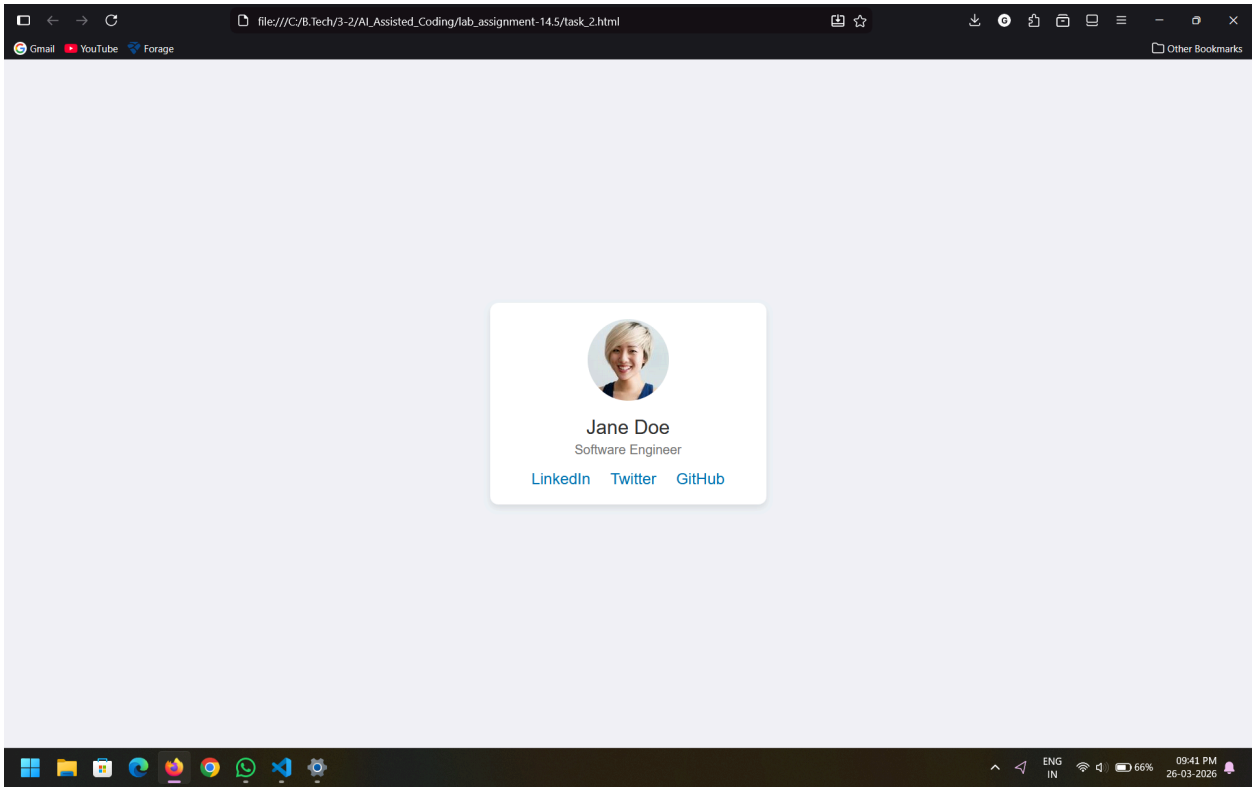
- Display user image, name, role, and social media links.
- Use AI to:
  - o Suggest modern UI colors and fonts.
  - o Design the card using Flexbox.
  - o Add hover animation and shadow effects

Copilot Suggested Code :

task_2.html > html

```html
<!— Create a profile card web page. —>

<!DOCTYPE html>
<html lang="en">
<head>
    <title>Profile Card</title>
    <style>
        body {
            font-family: 'Helvetica Neue', sans-serif;
            background-color: #f0f4f8;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
            margin: 0;
        }
        .profile-card {
            background-color: #ffffff;
            border-radius: 10px;
            box-shadow: 0 4px 8px rgba(0,0,0,0.1);
            width: 300px;
            padding: 20px;
            text-align: center;
            transition: transform 0.3s ease, box-shadow 0.3s ease;
        }
        .profile-card:hover {
            transform: translateY(-5px);
            box-shadow: 0 8px 16px rgba(0,0,0,0.2);
        }
        .profile-image {
            width: 100px;
            height: 100px;
            border-radius: 50%;
            object-fit: cover;
            margin-bottom: 15px;
        }
        .profile-name {
            font-size: 1.5em;
            color: #333333;
            margin-bottom: 5px;
        }
        .profile-role {
            font-size: 1em;
            color: #777777;
            margin-bottom: 15px;
        }
        .social-links a {
            color: #0077b5; /* LinkedIn Blue */
            text-decoration: none;
            margin: 0 10px;
            font-size: 1.2em;
        }
    </style>
</head>
<body>
    <div class="profile-card">
        <img src="https://randomuser.me/api/portraits/women/44.jpg" alt="Profile Image" class="profile-image">
        <div class="profile-name">Jane Doe</div>
        <div class="profile-role">Software Engineer</div>
        <div class="social-links">
            <a href="https://linkedin.com/in/jane-doe" target="_blank">LinkedIn</a>
            <a href="https://twitter.com/jane_doe" target="_blank">Twitter</a>
            <a href="https://github.com/jane-doe" target="_blank">GitHub</a>
        </div>
    </div>
</body>
</html>
```

Output :



Task 3 -Develop a dynamic shopping cart system.

Requirements:

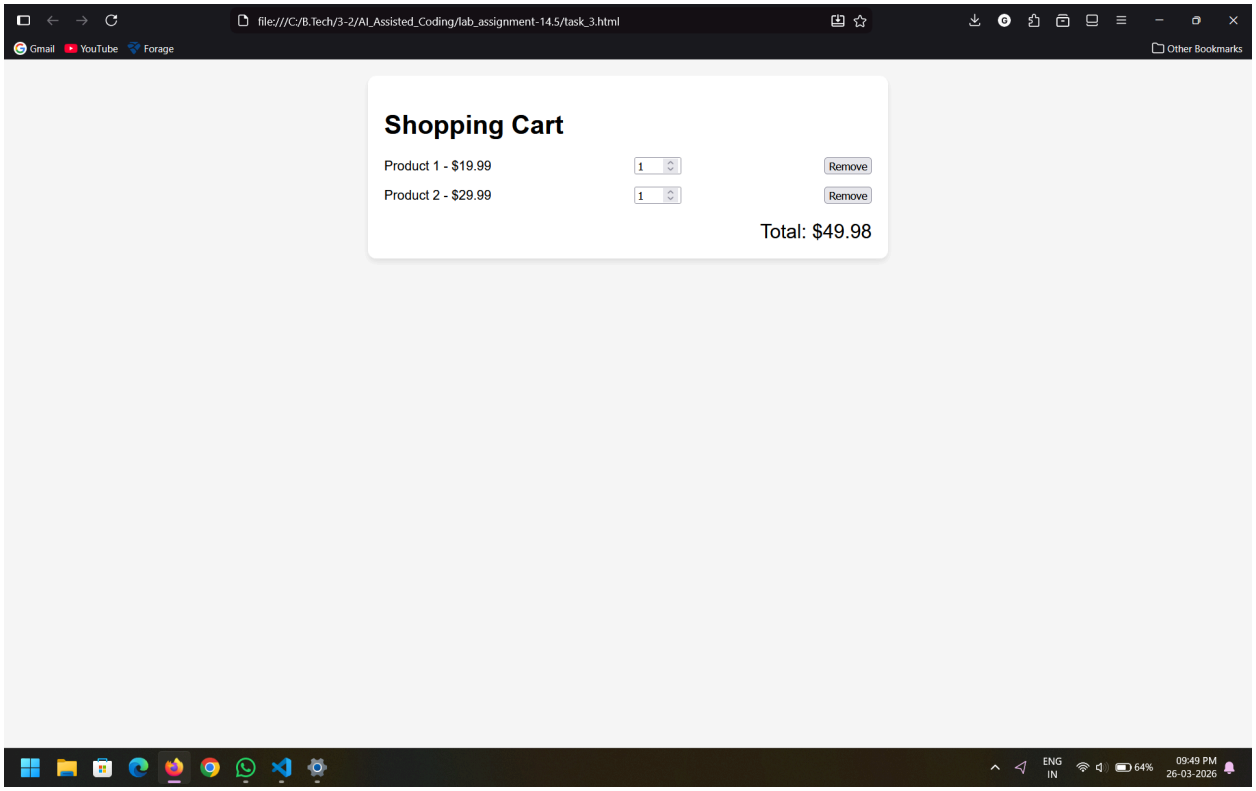
- Add, remove, and update product quantities.
- Display real-time total price.
- Use AI to:
    - o Manage cart state using JavaScript objects.
    - o Persist cart data in localStorage.

Copilot Suggested Code :

task_3.html > html

```html
<!— Task 3 -Develop a dynamic shopping cart system.
Requirements:
• Add, remove, and update product quantities.
• Display real-time total price.
• Use AI to:
o Manage cart state using JavaScript objects.
o Persist cart data in localStorage. —>

<!DOCTYPE html>
<html lang="en">
<head>
    <title>Shopping Cart</title>
    <style>
        body {
            font-family: 'Arial', sans-serif;
            margin: 0;
            padding: 20px;
            background-color: #f9f9f9;
        }
        .cart-container {
            max-width: 600px;
            margin: 0 auto;
            background-color: #fff;
            padding: 20px;
            border-radius: 10px;
            box-shadow: 0 4px 8px rgba(0,0,0,0.1);
        }
        .cart-item {
            display: flex;
            justify-content: space-between;
            align-items: center;
            margin-bottom: 15px;
        }
        .cart-item input {
            width: 50px;
        }
        .total-price {
            font-size: 1.5em;
            text-align: right;
            margin-top: 20px;
        }
    </style>
</head>
<body>
    <div class="cart-container">
        <h1>Shopping Cart</h1>
        <div id="cart-items"></div>
        <div class="total-price" id="total-price">Total: $0.00</div>
    </div>

    <script>
        const cart = JSON.parse(localStorage.getItem('cart')) || {};

        function updateCart() {
            const cartItemsContainer = document.getElementById('cart-items');
            cartItemsContainer.innerHTML = '';
            let totalPrice = 0;

            for (const productId in cart) {
                const item = cart[productId];
                totalPrice += item.price * item.quantity;

                const cartItem = document.createElement('div');
                cartItem.className = 'cart-item';
                cartItem.innerHTML = `
                    <span>${item.name} - $${item.price.toFixed(2)}</span>
                    <input type="number" value="${item.quantity}" min="1" onchange="updateQuantity('${productId}', this.value)">
                    <button onclick="removeFromCart('${productId}')">Remove</button>
                `;
                cartItemsContainer.appendChild(cartItem);
            }

            document.getElementById('total-price').textContent = `Total: $${totalPrice.toFixed(2)}`;
            localStorage.setItem('cart', JSON.stringify(cart));
        }

        function addToCart(product) {
            if (cart[product.id]) {
                cart[product.id].quantity += 1;
            } else {
                cart[product.id] = { ...product, quantity: 1 };
            }
            updateCart();
        }

        function updateQuantity(productId, quantity) {
            if (cart[productId]) {
                cart[productId].quantity = parseInt(quantity);
                updateCart();
            }
        }

        function removeFromCart(productId) {
            delete cart[productId];
            updateCart();
        }

        // Example products
        addToCart({ id: '1', name: 'Product 1', price: 19.99 });
        addToCart({ id: '2', name: 'Product 2', price: 29.99 });
    </script>
</body>
</html>
```

Output :



Task 4 – Role-Based Access Control Application

Create a role-based web application.

Requirements:

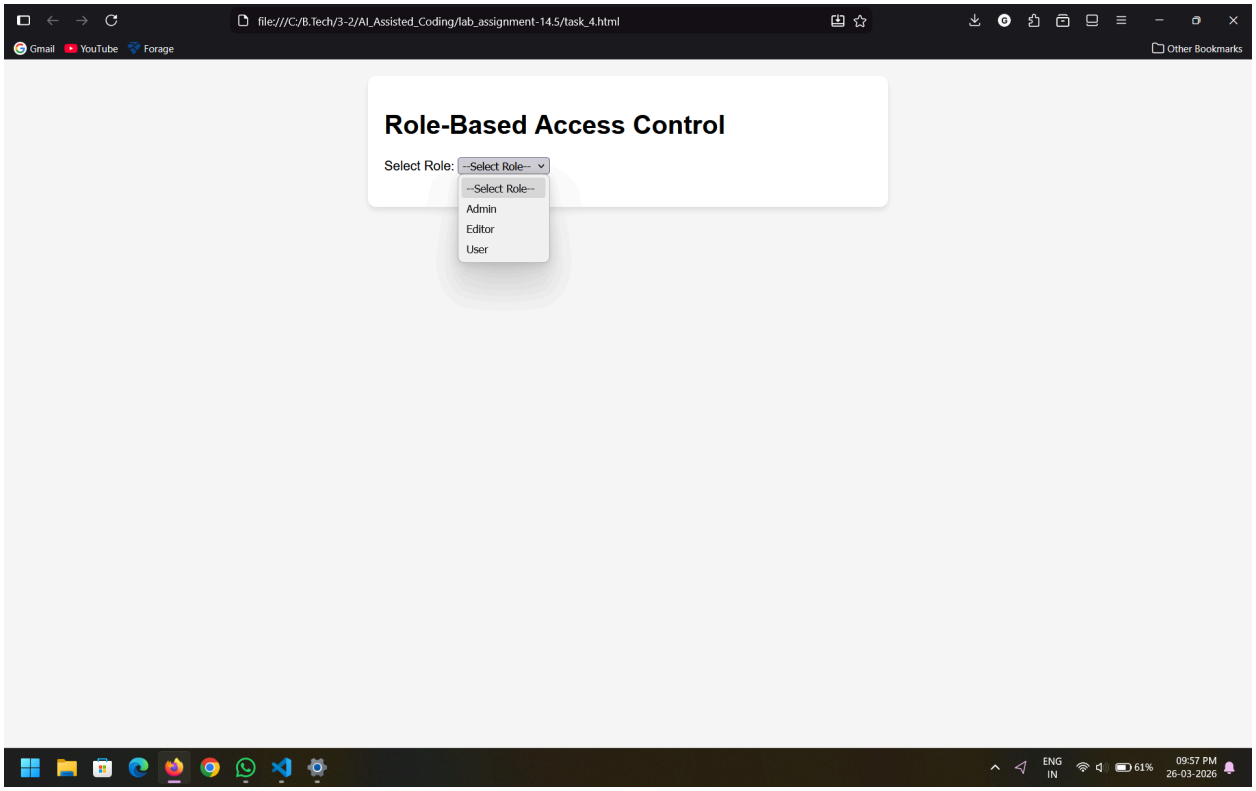
- Admin, Editor, User roles.
- Conditional UI rendering.
- Use AI to:
  - o Design access control logic.
  - o Maintain secure state transitions.
  - o Create scalable UI components.

Copilot Suggested Code :

task_4.html > ...

```html
<!— Create a role-based web application. —>

<!DOCTYPE html>
<html lang="en">
<head>
    <title>Role-Based Access Control</title>
    <style>
        body {
            font-family: 'Arial', sans-serif;
            margin: 0;
            padding: 20px;
            background-color: #f9f9f9;
        }
        .container {
            max-width: 600px;
            margin: 0 auto;
            background-color: #fff;
            padding: 20px;
            border-radius: 10px;
            box-shadow: 0 4px 8px rgba(0,0,0,0.1);
        }
        .role-select {
            margin-bottom: 20px;
        }
        .content {
            display: none;
        }
        .content.active {
            display: block;
        }
    </style>
</head>
<body>
    <div class="container">
        <h1>Role-Based Access Control</h1>
        <div class="role-select">
            <label for="role">Select Role:</label>
            <select id="role" onchange="renderContent()">
                <option value="">--Select Role--</option>
                <option value="admin">Admin</option>
                <option value="editor">Editor</option>
                <option value="user">User</option>
            </select>
        </div>
        <div id="admin-content" class="content">
            <h2>Admin Dashboard</h2>
            <p>Welcome, Admin! You have full access to the system.</p>
        </div>
        <div id="editor-content" class="content">
            <h2>Editor Dashboard</h2>
            <p>Welcome, Editor! You can edit content but have limited access to settings.</p>
        </div>
        <div id="user-content" class="content">
            <h2>User Dashboard</h2>
            <p>Welcome, User! You can view content but cannot make changes.</p>
        </div>
    </div>

    <script>
        function renderContent() {
            const role = document.getElementById('role').value;
            document.querySelectorAll('.content').forEach(content => {
                content.classList.remove('active');
            });
            if (role) {
                document.getElementById(`${role}-content`).classList.add('active');
            }
        }

```

Output :



Task 5 – Multi-Language Website

Design a multilingual website.

Requirements:

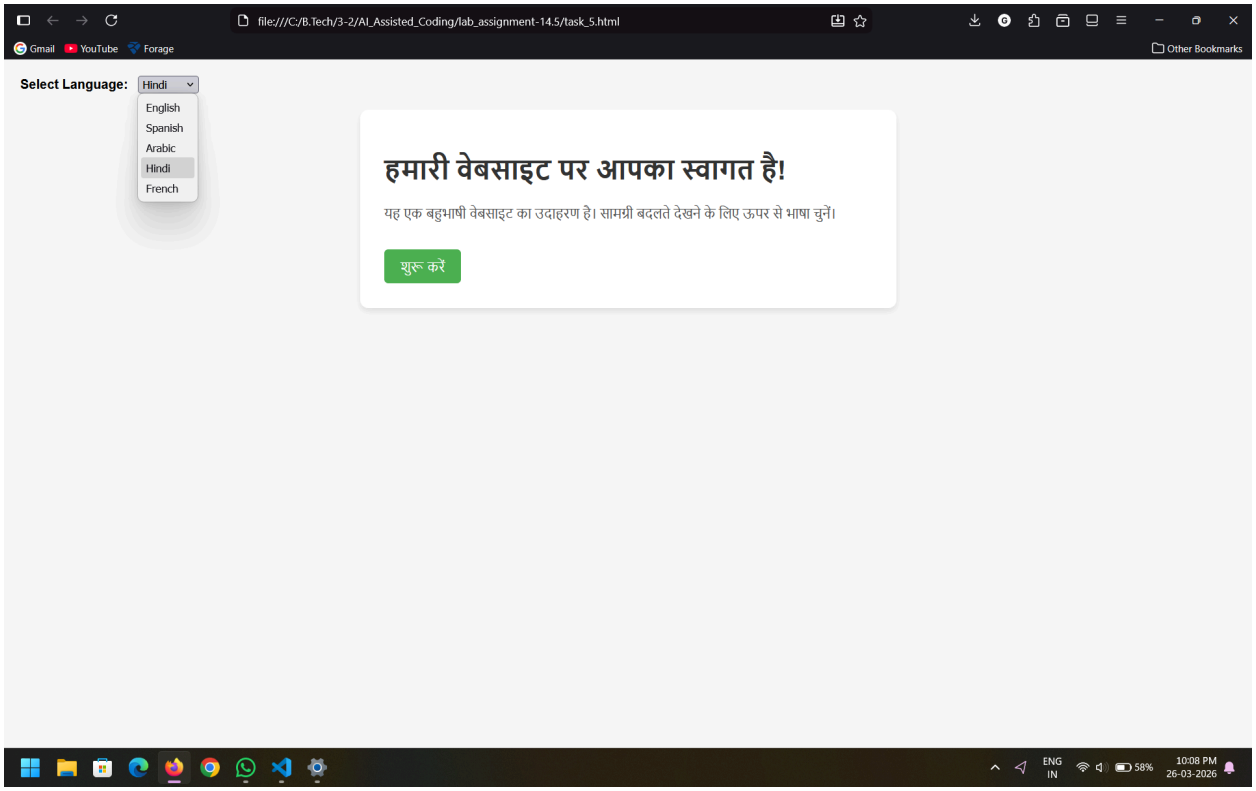
- Switch languages dynamically.
- Persist language preference.
- Use AI to:
    - o Structure translation files.
    - o Handle RTL/LTR layouts.
    - o Improve accessibility.

Copilot Suggested Code :

task_5.html > html

```html
<!-- Design a multilingual website. -->
<!DOCTYPE html>
<html lang="en">

<head>
    <title>Multi-Language Website</title>
    <style>
        body {
            font-family: 'Arial', sans-serif;
            margin: 0;
            padding: 20px;
            background-color: #f9f9f9;
        }

        .language-select {
            margin-bottom: 20px;
        }

        .language-select label {
            font-weight: bold;
            margin-right: 8px;
        }

        .content {
            max-width: 600px;
            margin: 0 auto;
            background-color: #fff;
            padding: 30px;
            border-radius: 10px;
            box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
        }

        .content h1 {
            color: #333;
        }

        .content p {
            color: #555;
            line-height: 1.6;
        }

        .content button {
            margin-top: 15px;
            padding: 10px 20px;
            background-color: #4CAF50;
            color: white;
            border: none;
            border-radius: 5px;
            cursor: pointer;
            font-size: 16px;
        }

        .content button:hover {
            background-color: #45a049;
        }
    </style>
</head>

<body>
    <div class="language-select">
        <label for="language">Select Language:</label>
        <select id="language" onchange="switchLanguage()">
            <option value="en">English</option>
            <option value="es">Spanish</option>
            <option value="ar">Arabic</option>
            <option value="hi">Hindi</option>
            <option value="fr">French</option>
        </select>
    </div>
    <div class="content" id="content">
        <!-- Content will be dynamically updated based on language selection -->
    </div>

    <script>
        const translations = {
            en: {
                welcome: "Welcome to our Website!",
                description: "This is a multilingual website example. Select a language from the dropdown above to see the content change.",
                btn: "Get Started"
            },
            es: {
                welcome: "¡Bienvenido a nuestro sitio web!",
                description: "Este es un ejemplo de sitio web multilingüe. Seleccione un idioma del menú desplegable para ver el contenido cambiar.",
                btn: "Empezar"
            },
            ar: {
                welcome: "مرحبًا بكم في موقعنا!",
                description: "هذا مثال على موقع ويب متعدد اللغات. اختر لغة من القائمة المنسدلة لترى تغيير المحتوى.",
                btn: "ابدأ"
            },
            hi: {
                welcome: "हमारी वेबसाइट पर आपका स्वागत है!",
                description: "यह एक बहुभाषी वेबसाइट का उदाहरण है। सामग्री बदलते देखने के लिए ऊपर से भाषा चुनें।",
                btn: "शुरू करें"
            },
            fr: {
                welcome: "Bienvenue sur notre site web!",
                description: "Ceci est un exemple de site web multilingue. Sélectionnez une langue dans le menu déroulant pour voir le contenu changer.",
                btn: "Commencer"
            }
        };

        function switchLanguage() {
            const language = document.getElementById('language').value;
            localStorage.setItem('preferredLanguage', language);
            renderContent(language);
        }

        function renderContent(language) {
            const content = document.getElementById('content');
            const t = translations[language];
            content.innerHTML = `
                <h1>${t.welcome}</h1>
                <p>${t.description}</p>
                <button>${t.btn}</button>
            `;
            document.body.dir = language === 'ar' ? 'rtl' : 'ltr';
            document.documentElement.lang = language;
        }

        // Load preferred language on page load
        const preferredLanguage = localStorage.getItem('preferredLanguage') || 'en';
        document.getElementById('language').value = preferredLanguage;
        renderContent(preferredLanguage);
    </script>
</body>
```

Output :



Task 6 – Online Examination System

Create an online exam interface.

Requirements:

- Timer, questions, and submission.
- Prevent multiple submissions.
- Use AI to:
    - o Manage exam state.
    - o Improve exam UX.
    - o Ensure accessibility.

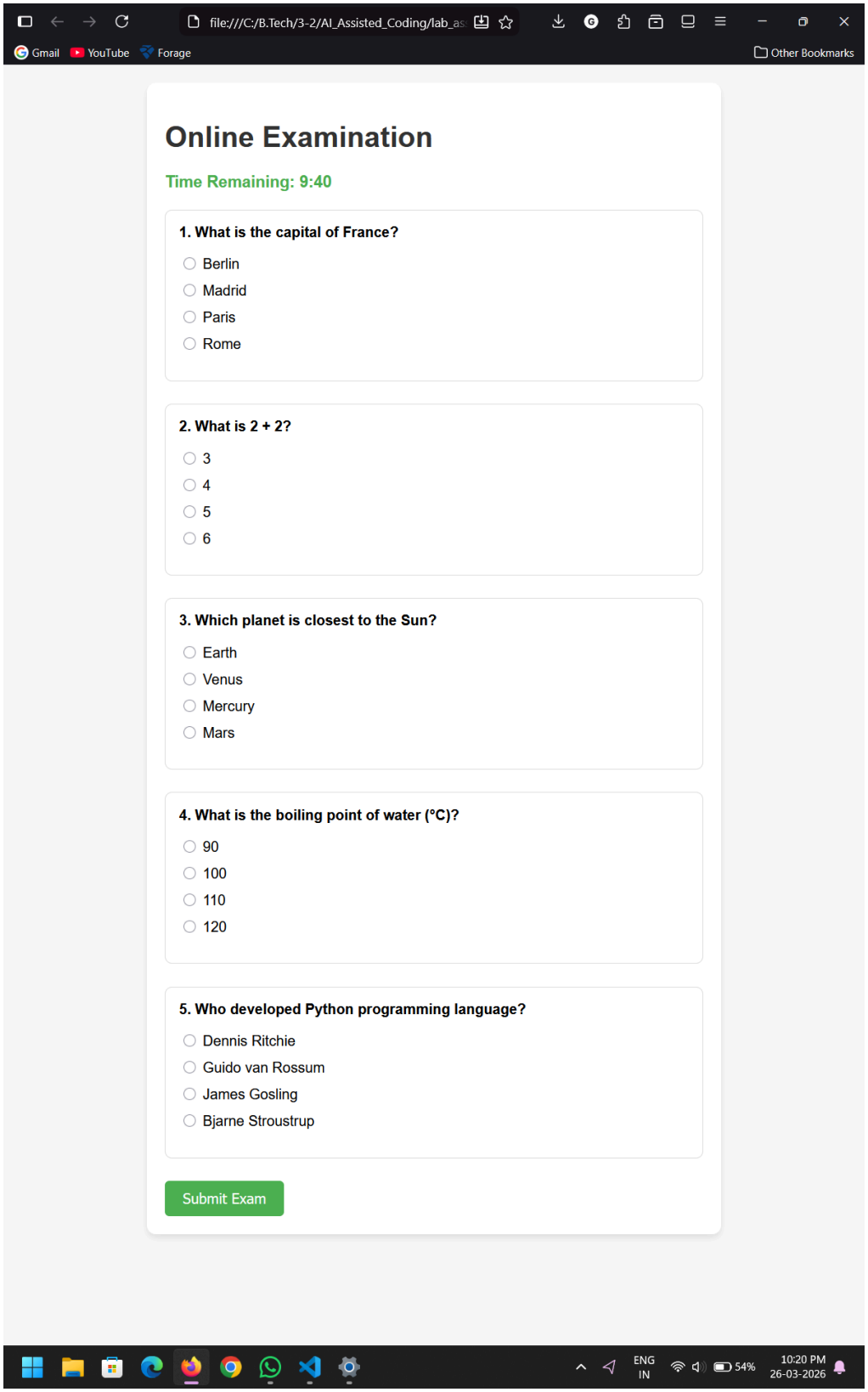
Copilot Suggested Code :

task_6.html > html > body

```html
<!— Task 6 - Online Examination System —>
<!DOCTYPE html>
<html lang="en">

<head>
    <title>Online Examination System</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 20px;
            background-color: #f9f9f9;
        }

        .exam-container {
            max-width: 600px;
            margin: 0 auto;
            background-color: #fff;
            padding: 20px;
            border-radius: 10px;
            box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
        }

        h1 {
            color: #333;
        }

        #timer {
            font-size: 18px;
            font-weight: bold;
            color: #4CAF50;
            margin-bottom: 20px;
        }

        #timer.warning {
            color: orange;
        }

        #timer.danger {
            color: red;
        }

        .question {
            margin-bottom: 25px;
            padding: 15px;
            border: 1px solid #ddd;
            border-radius: 8px;
        }

        .question p {
            font-weight: bold;
            margin-top: 0;
        }

        .options {
            list-style-type: none;
            padding: 0;
        }

        .options li {
            margin-bottom: 10px;
        }

        .options label {
            cursor: pointer;
        }

        .submit-btn {
            padding: 10px 20px;
            background-color: #4CAF50;
            color: white;
            border: none;
            border-radius: 5px;
            cursor: pointer;
            font-size: 16px;
        }

        .submit-btn:hover {
            background-color: #45a049;
        }

        .submit-btn:disabled {
            background-color: #aaa;
            cursor: not-allowed;
        }

        #result {
            margin-top: 20px;
            padding: 15px;
            background-color: #e8f5e9;
            border-radius: 8px;
            display: none;
        }
    </style>
</head>
```

```html
<body>
    <div class="exam-container">
        <h1>Online Examination</h1>
        <div id="timer">Time Remaining: 10:00</div>
        <div id="questions"></div>
        <button class="submit-btn" id="submitBtn" onclick="submitExam()">Submit Exam</button>
        <div id="result"></div>
    </div>

    <script>
        const examState = {
            timeRemaining: 600,
            questions: [
                { id: 1, text: "What is the capital of France?", options: ["Berlin", "Madrid", "Paris", "Rome"], answer: 2 },
                { id: 2, text: "What is 2 + 2?", options: ["3", "4", "5", "6"], answer: 1 },
                { id: 3, text: "Which planet is closest to the Sun?", options: ["Earth", "Venus", "Mercury", "Mars"], answer: 2 },
                { id: 4, text: "What is the boiling point of water (°C)?", options: ["90", "100", "110", "120"], answer: 1 },
                { id: 5, text: "Who developed Python programming language?", options: ["Dennis Ritchie", "Guido van Rossum", "James Gosling", "Bjarne Stroustrup"]
            ],
            answers: {},
            submitted: false
        };

        let timerInterval;

        function startTimer() {
            const timerElement = document.getElementById('timer');
            timerInterval = setInterval(() => {
                if (examState.timeRemaining <= 0) {
                    clearInterval(timerInterval);
                    submitExam();
                } else {
                    examState.timeRemaining--;
                    const minutes = Math.floor(examState.timeRemaining / 60);
                    const seconds = examState.timeRemaining % 60;
                    timerElement.textContent = `Time Remaining: ${minutes}:${seconds < 10 ? '0' : ''}${seconds}`;

                    if (examState.timeRemaining <= 60) {
                        timerElement.className = 'danger';
                    } else if (examState.timeRemaining <= 180) {
                        timerElement.className = 'warning';
                    }
                }
            }, 1000);
        }

        function renderQuestions() {
            const questionsContainer = document.getElementById('questions');
            examState.questions.forEach((question, qIndex) => {
                const questionElement = document.createElement('div');
                questionElement.className = 'question';
                questionElement.setAttribute('role', 'group');
                questionElement.setAttribute('aria-labelledby', `q${question.id}`);
                questionElement.innerHTML = `
                    <p id="q${question.id}">${qIndex + 1}. ${question.text}</p>
                    <ul class="options">
                        ${question.options.map((option, index) => `
                            <li>
                                <label>
                                    <input type="radio" name="question-${question.id}" value="${index}"
                                        onchange="selectAnswer(${question.id}, ${index})"
                                        aria-label="${option}">
                                    ${option}
                                </label>
                            </li>
                        `).join('')}
                    </ul>
                `;
                questionsContainer.appendChild(questionElement);
            });
        }

        function selectAnswer(questionId, optionIndex) {
            examState.answers[questionId] = optionIndex;
        }

        function submitExam() {
            if (examState.submitted) {
                alert("You have already submitted the exam!");
                return;
            }

            clearInterval(timerInterval);
            examState.submitted = true;

            document.querySelectorAll('input[type="radio"]').forEach(input => input.disabled = true);
            document.getElementById('submitBtn').disabled = true;
            document.getElementById('timer').textContent = "Exam Submitted!";

            // Calculate score
            let score = 0;
            examState.questions.forEach(q => {
                if (examState.answers[q.id] === q.answer) score++;
            });

            const total = examState.questions.length;
            const percent = Math.round((score / total) * 100);

            const resultDiv = document.getElementById('result');
            resultDiv.style.display = 'block';
            resultDiv.innerHTML = `
                <h2>Exam Results</h2>
                <p>You scored <strong>${score} / ${total}</strong> (${percent}%)</p>
                <p>${percent >= 60 ? 'Congratulations! You passed!' : 'You did not pass. Better luck next time!'}</p>
            `;

            resultDiv.scrollIntoView({ behavior: 'smooth' });
        }

        renderQuestions();
        startTimer();
    </script>
</body>
</html>
```

Output :


file:///C:/B.Tech/3-2/AI_Assisted_Coding/lab_as
Gmail
YouTube
Forage
Other Bookmarks
Online Examination
Time Remaining: 9:40
1. What is the capital of France?
Berlin
Madrid
Paris
Rome
2. What is 2 + 2?
3
4
5
6
3. Which planet is closest to the Sun?
Earth
Venus
Mercury
Mars
4. What is the boiling point of water (°C)?
90
100
110
120
5. Who developed Python programming language?
Dennis Ritchie
Guido van Rossum
James Gosling
Bjarne Stroustrup
Submit Exam
ENG
IN
54%
10:20 PM
26-03-2026